Yujuan Xie, Ph.D. Senior iOS Developer / Mobile Software Engineer

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PROFESSIONAL SUMMARY

Accomplished Senior iOS Developer with a Ph.D. analytical background and proven track record of architecting, building, and publishing **16+ commercial iOS applications on the Apple App Store** (Florithm LLC). Combines deep engineering expertise in Swift, SwiftUI, UIKit, Combine, StoreKit 2 monetization, Core ML/Vision, and real-time cloud architectures (Firebase) with doctoral-level computational modeling, algorithmic problem solving, and rigorous data analysis. Expert in 0-to-1 mobile lifecycle execution, 120Hz ProMotion UI optimization, and strict COPPA privacy compliance.

TECHNICAL SKILLS

iOS & Languages:	Swift, SwiftUI, UIKit, Combine, Objective-C, TypeScript, JavaScript (ES6+), HTML5/CSS3, Python, C/C++
Apple Frameworks:	StoreKit 2 (In-App Purchases & Subscriptions), Core Data, Vision, Core ML, UserNotifications (APNs), LocalAuthentication (Face ID), AVFoundation, Security / Keychain, GCD, Swift Concurrency (async/await, Actors)
Architecture & Patterns:	MVVM, Coordinator, Clean Architecture, Reactive Programming, Offline-First / Caching, Native Bridge Plugins (Capacitor/React Native)
Cloud & Backend:	Firebase (Cloud Firestore, Functions, Auth, Storage), Real-Time WebSockets, RESTful APIs, Node.js
Tooling & Scientific Computing:	Xcode, Instruments (Time Profiler, Allocations, Leaks), TestFlight, App Store Connect, Git/GitHub Actions, XCTest, Statistical Modeling

ENGINEERING & RESEARCH EXPERIENCE

Florithm LLC — Founder & Principal iOS Developer

2019 – Present | Fremont, CA

- **End-to-End iOS Product Delivery:** Architected, built, and launched 16 production iOS apps on the Apple App Store across FinTech, Productivity, Education, and Lifestyle, achieving rapid release cycles and 99.9% crash-free sessions.
- **StoreKit 2 & Monetization Architecture:** Integrated native StoreKit 2 workflows for auto-renewable subscriptions, consumables, and non-consumable lifetime passes with automated receipt verification and entitlement tracking.
- **Real-Time Offline-First Sync:** Built high-efficiency mobile data pipelines using Cloud Firestore, local caching, and optimistic UI mutations for instant response under intermittent network conditions.
- **Memory Profiling & Performance:** Leveraged Xcode Instruments (Allocations, Leaks, Time Profiler) to eliminate ARC retain cycles, minimize CPU overhead, and guarantee smooth 120Hz ProMotion scrolling.
- **Security & Child Privacy (COPPA):** Designed zero-friction 6-digit cryptographic hardware pairing protocols enabling children to sync devices without collecting emails, phone numbers, or passwords.
- **Hardware & Media Integration:** Built biometric Face ID authentication via LocalAuthentication, secure Keychain storage, camera pipelines with AVFoundation, and Core ML/Vision client-side color analysis.

4D Labs, Simon Fraser University — Research Scientist & Quantitative Data Analyst

2008 – 2013 | Burnaby, BC, Canada

- Applied advanced statistical data analysis, mathematical modeling, and experimental matrices to investigate structural phase transitions and material reliability under Office of Naval Research-funded programs.
- Authored 3 peer-reviewed scientific publications in premier journals (Physical Review Letters, IEEE Transactions) and won 1st Place at the IEEE International Symposium on Applications of Ferroelectrics.

FEATURED IOS PRODUCTION APPS & ARCHITECTURAL HIGHLIGHTS

The Family Bank — Kids Money & Allowance Bank

Finance · EdTech · Cloud Sync

Virtual banking simulation teaching kids compound interest, chore bounties, and financial habits.

- **Automated Compound Interest Engine:** Daily interest calculation algorithms crediting ledgers based on parent APY.
- **6-Digit Device Pairing:** Multi-device cryptographic pairing linking child hardware directly without credentials (COPPA compliant).
- **Chore Approvals & StoreKit:** Built real-time parent verification workflow and integrated StoreKit lifetime family unlock.

PlanCrew — Group Chat & Planning

Social · Productivity

Collaborative event planning and group chat platform for crews.

- Real-time chat with threaded replies, reactions, and media caching.
- Interactive calendar event planning, itineraries, and shared checklists.

My Color Palette: AI Stylist

Fashion Tech · Vision AI

AI-driven color analysis categorizing users into 12 seasonal palettes.

- Client-side colorimetric analysis processing skin tone and undertones.
- Custom palette visualizers and offline-ready wardrobe tools.

Nexus — Career Discovery

Education · Career

Psychometric orientation tool utilizing the Holland Code (RIASEC).

- 36-question scoring engine mapping personality traits to career paths.
- Animated radar charts and personality breakdown screens.

Sparking Joy & TapTapTask

Lifestyle · Gamification

Mindful decluttering companion & family task rewards systems.

- Category-by-category workflows, milestone rewards, and streaks.
- Gamified task rewards with customizable goal progress meters.

APP STORE PORTFOLIO (16 PUBLISHED IOS APPS)

Domain	Production iOS Apps on Apple App Store
Finance & Family	**The Family Bank** (Kids Money & Allowance Bank)
Productivity & Tools	ScheduleAround, TapTapTask, Countdowns (Visual Timers), Intervals (Count-up Timer), Plateful+
Fashion & Creative	My Color Palette: AI Stylist, Profile Picture Maker: Avatars
Education & Literacy	Nexus (Career Discovery), RightRead (K-6 Leveled Books), Phonics Cards (5,000+ words), Read with Booki
Social & Games	PlanCrew (Group Chat & Plans), Fleet Keeper+ (Space Arcade), Fish or Bait (Retro Survival), Am I a Narcissist?

EDUCATION

Ph.D., Chemistry — Simon Fraser University, Canada (2013)

M.Eng., Materials Science — Hefei University of Technology, China (2008)

B.Eng., Materials Engineering — Hefei University of Tech (2006, Top 5%)